

Networks and Systems Management



Software-Defined Networking (SDN)

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What is SDN?

- Traditional networks: tightly coupled control & data planes.
- SDN: An approach to networking that separates control plane from data plane.
- Centralized control through a software-based controller.





Why SDN?



Complexity of traditional networks.



Slow response to network changes.



Lack of flexibility and programmability.

SDN Architecture Overview



Application Layer: Network applications (firewalls, load balancers).



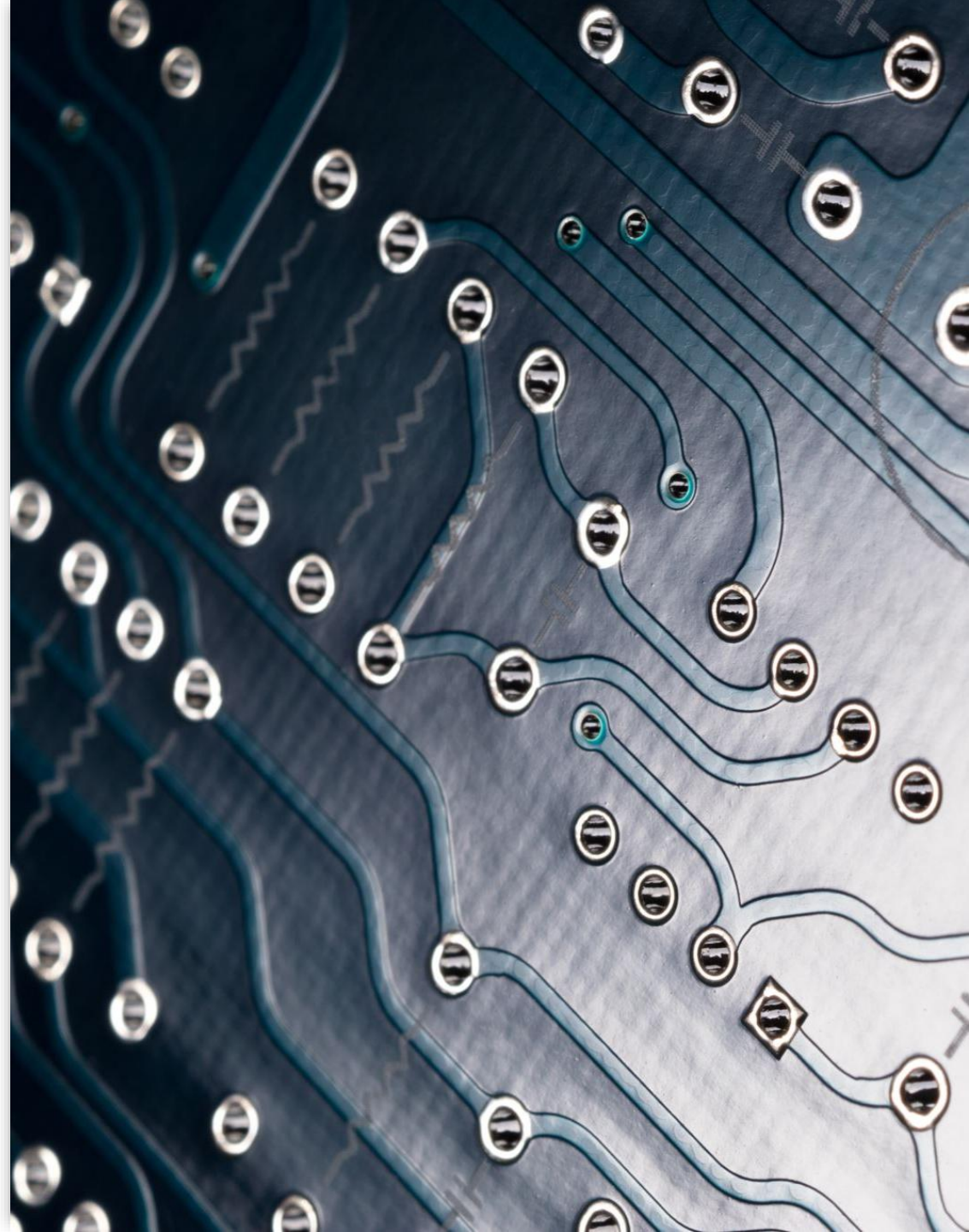
Control Layer: SDN controller (e.g., OpenDaylight, ONOS).



Infrastructure Layer: Network devices (switches, routers).

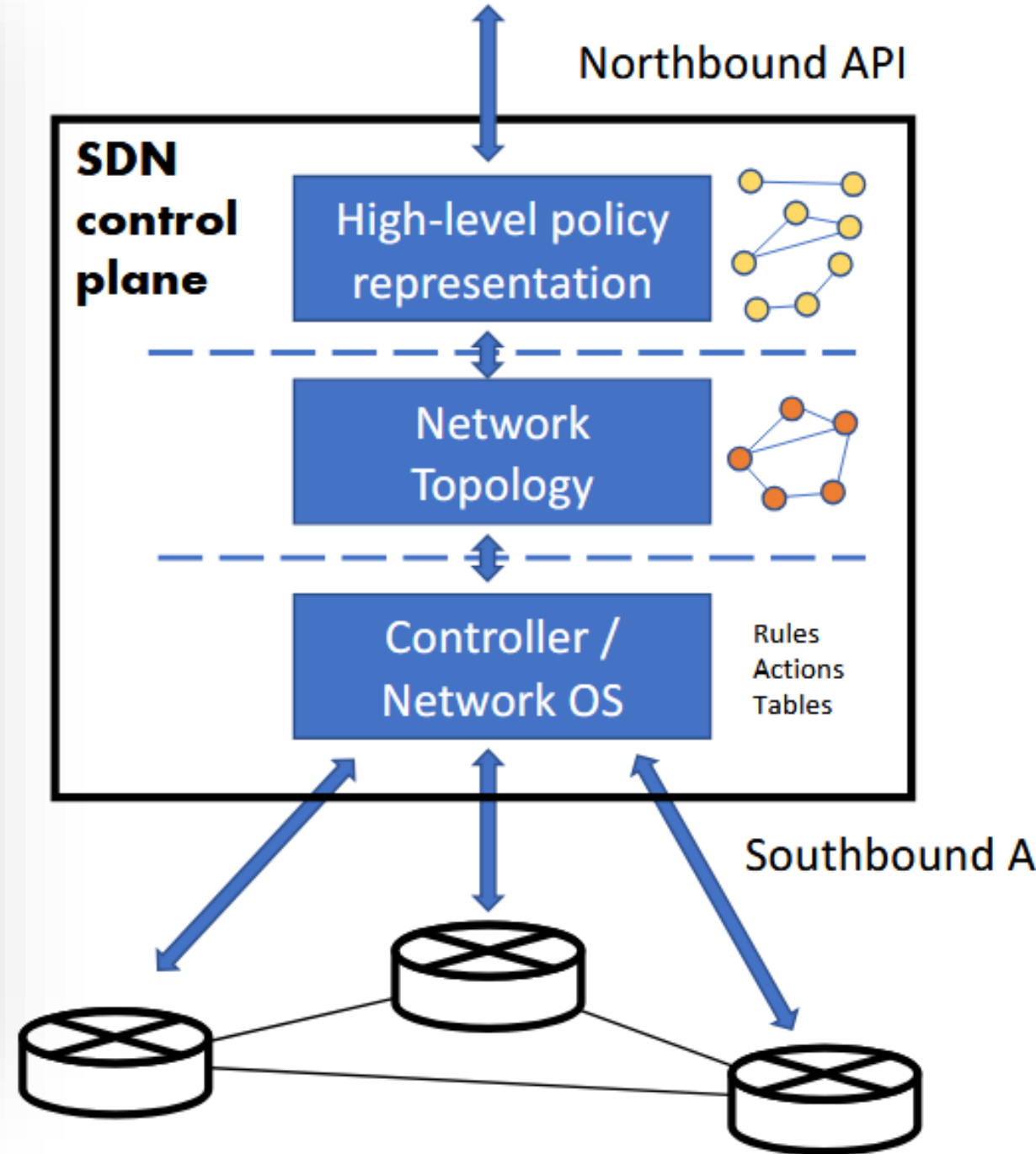
SDN Components

- Controller: The brain of the network (e.g., Ryu, Floodlight).
- Southbound API: Protocols like OpenFlow.
- Northbound API: Interfaces to applications.
- Data Plane Devices: Forward packets as instructed by the controller.



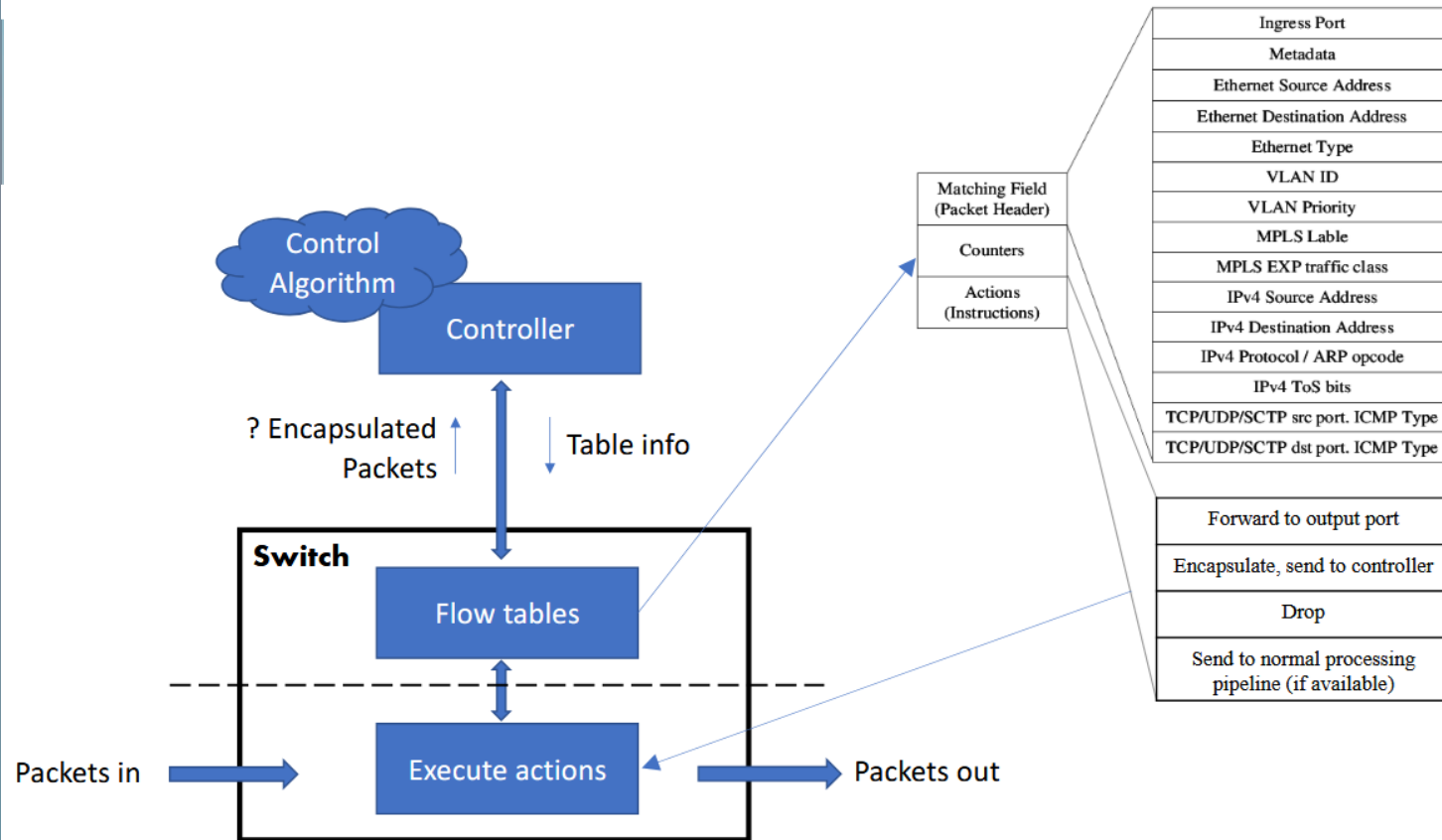
Scaling up

- Control multiple switches
- Abstract network topology
- Support high-level
 - representation of policies
 - (e.g. virtualization, intents)



OpenFlow Protocol

- Introduced by Stanford in 2008.
- Allows controller to communicate with forwarding devices.
- Flow table entries, actions, and priorities.



Access Control Example

- Access control
 - 10.0.0.1 can ping 10.0.0.2
- Routing
 - Routing algorithm sets up destination address and forwarding port
 - Destination IP address – matching field
 - Forward to port – action

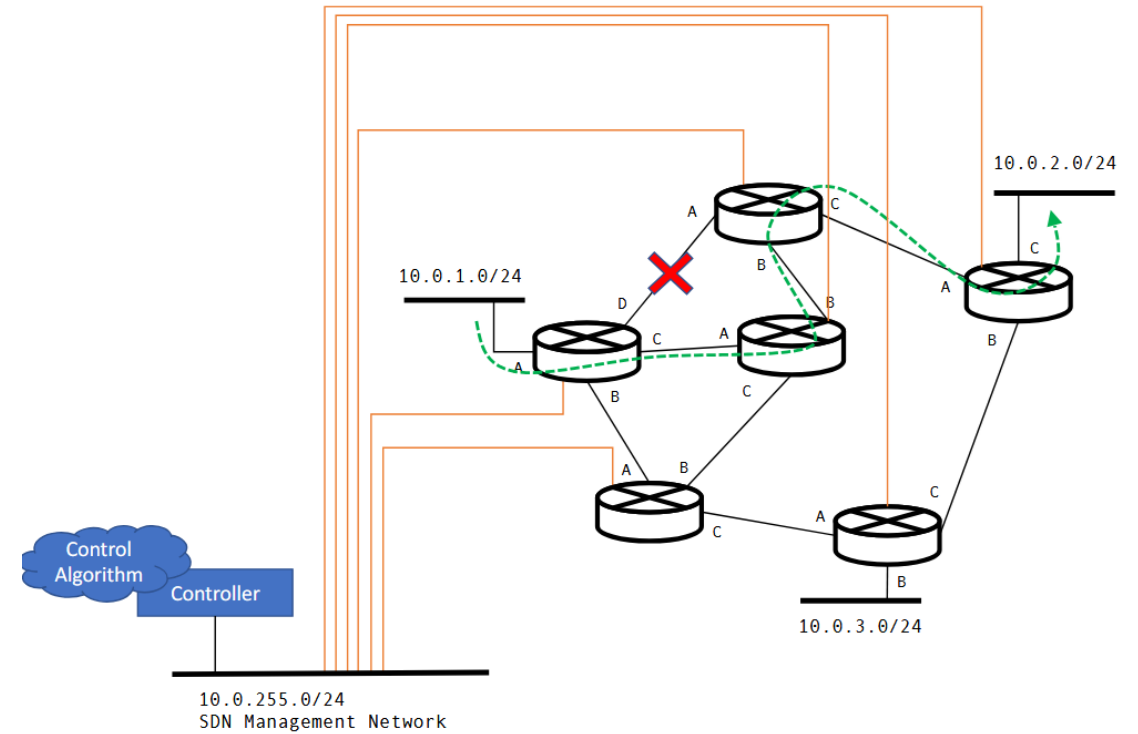
Matching fields			Default: drop
Source	Destination	Protocol	Action
10.0.0.1/32	10.0.0.2/32	ICMP	Forward port 2
10.0.0.2/32	10.0.0.1/32	ICMP	Forward port 1

SDN vs Traditional Networking

Feature	Traditional	SDN
Control	Distributed	Centralized
Configuration	Manual	Automated
Flexibility	Low	High
Innovation	Slow	Fast

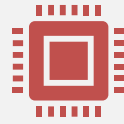
Benefits of SDN

- Centralized network control.
- Dynamic and programmable.
- Simplified network management.
- Faster innovation & deployment.





Challenges of SDN



Scalability of controllers.



Security concerns (e.g., controller as a target).



Interoperability with legacy systems.



Need for new skillsets.

SDN Use Cases

- Data center management.
- Network slicing in 5G.
- Traffic engineering.
- IoT and Edge computing.
- WAN optimization (e.g., Google B4).



B4 – Google's private WAN (2014)



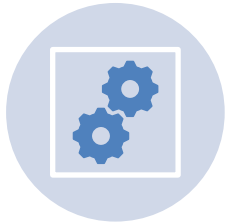
MERCHANT SILICON,
LOWER RELIABILITY



CONTROL PLANE IN
SOFTWARE



TARGET 100% LINK
UTILIZATION



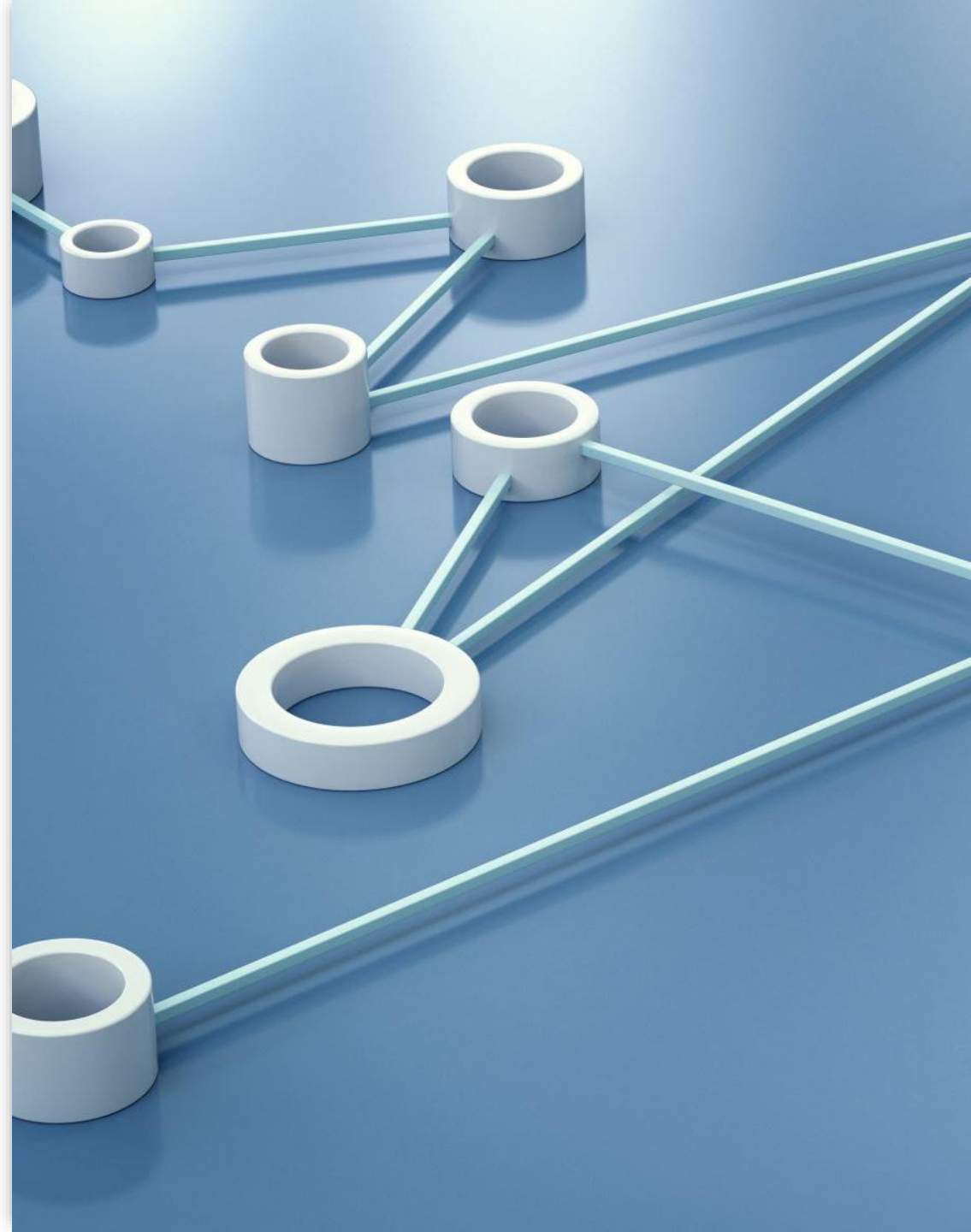
CENTRALIZED TRAFFIC
ENGINEERING



Sushant Jain, Alok Kumar, Subhasree Mandal, Joon Ong, Leon Poutievski, Arjun Singh, Subbaiah Venkata, Jim Wanderer, Junlan Zhou, Min Zhu, Jon Zolla, Urs Hölzle, Stephen Stuart, and Amin Vahdat. 2013. B4: experience with a globally-deployed software defined wan. In Proceedings of the ACM SIGCOMM 2013 conference on SIGCOMM (SIGCOMM '13). Association for Computing Machinery, New York, NY, USA, 3–14. DOI: <https://doi.org/10.1145/2486001.2486019>

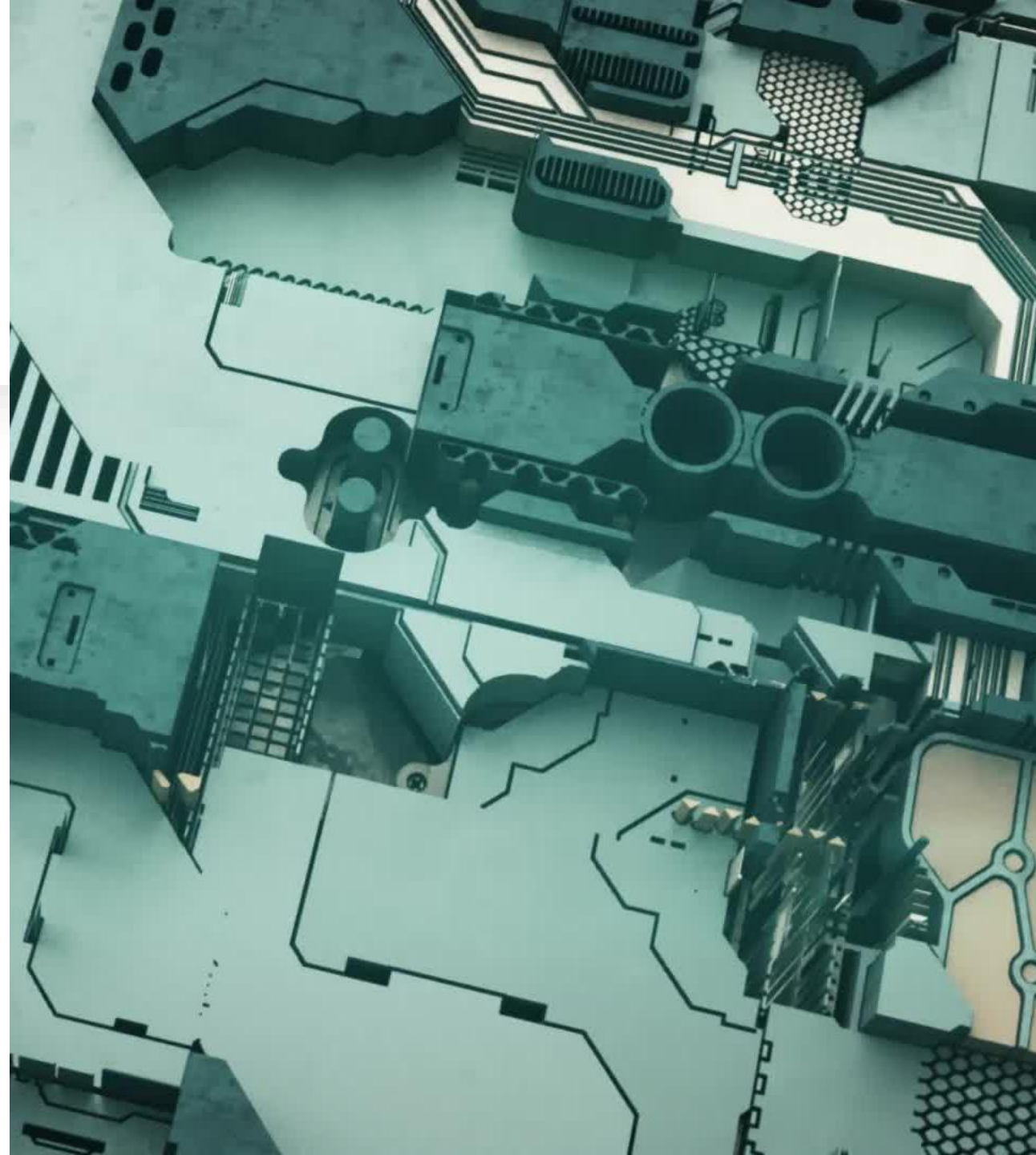
SDN and Network Management

- Real-time monitoring & telemetry.
- Automated provisioning and configuration.
- Policy-based management.
- Integration with orchestration tools (e.g., Kubernetes, Ansible).



SDN Tools and Platforms

- Controllers: OpenDaylight, ONOS, Ryu.
- Simulators/Emulators: Mininet, NS-3.
- Languages: P4 (for programming the data plane).

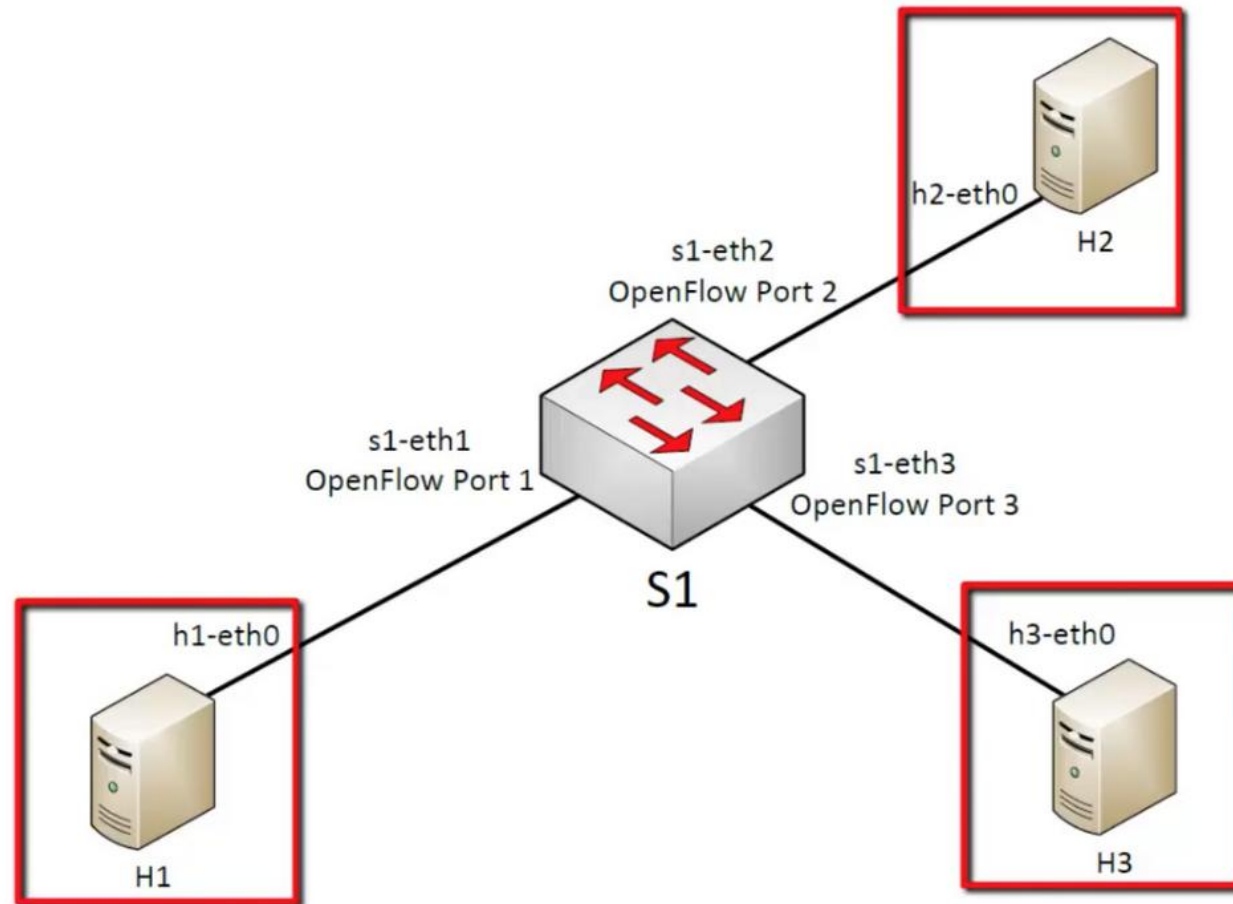


Future of SDN

- Integration with AI/ML.
- Intent-based networking.
- SD-WAN evolution.
- Programmable data planes with P4.

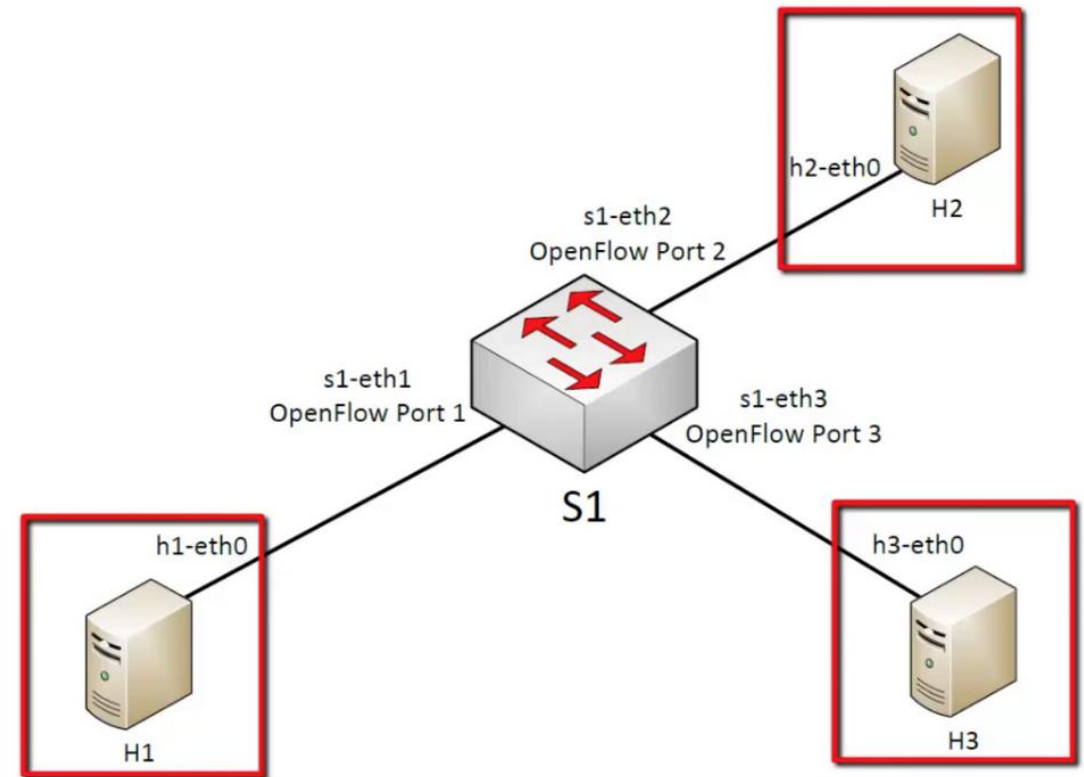


Example



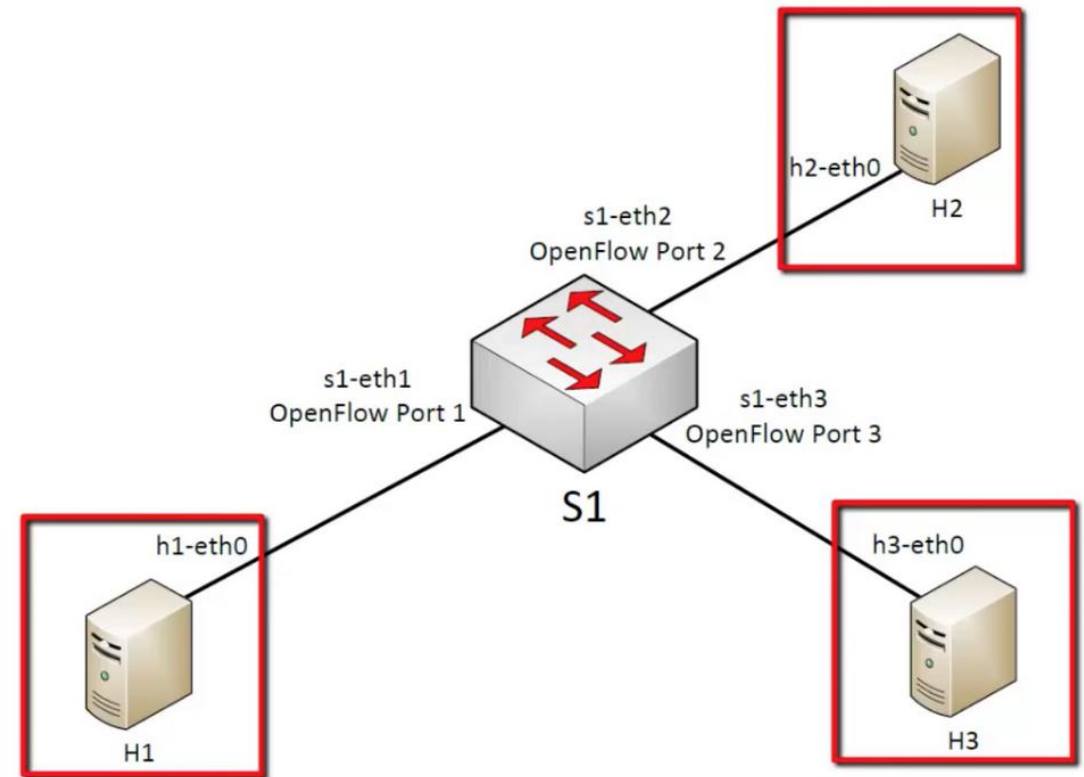
Example

- Setup docker with openvswitch and ryu
 - `docker run -it --name ryu --rm --privileged -v /lib/modules:/lib/modules osrg/ryu-book`
 - `docker exec -it ryu /bin/bash`
 - `root@6605a585b230:~# mn --topo=single,3 --controller=remote -mac`
 - `mininet> sh ovs-ofctl show s1`
 - `mininet> sh ovs-ofctl add-flow s1 action=normal`
 - `mininet> pingall`
 - `mininet> sh ovs-ofctl del-flows s1`
- <https://hackmd.io/@pmanzoni/SyWm3n0HH>



Example

- Start controller
 - `docker exec -it ryu /bin/bash`
 - `root@6605a585b230:~# mryu-manager --verbose ryu.app.ofctl_rest`
- Use restAPI to configure flows
 - `docker exec -it ryu /bin/bash`
 - `curl -X POST -d '{"dpid": 1,"match": {"in_port": 1},"actions": [{"type": "OUTPUT", "port": 2}]}' http://localhost:8080/stats/flowentry/add`
 - `curl -X POST -d '{"dpid": 1,"match": {"in_port": 2},"actions": [{"type": "OUTPUT", "port": 1}]}' http://localhost:8080/stats/flowentry/add`
 - `curl -X GET http://localhost:8080/stats/flow/1`
- Test in mininet the ping
 - `mininet> h1 ping h2`



Key Takeaways

SDN decouples control and data planes.

Provides agility, automation, and programmability.

Plays a key role in modern network management.

Further Reading & References

- Open Networking Foundation.
<https://opennetworking.org>
- Nick Feamster et al., 'The Road to SDN,'
ACM Queue, 2013.
- OpenFlow Switch Specification, ONF.
- Mininet: <http://mininet.org>
- ONOS Project: <https://onosproject.org>

